

1. Administrative & Study Identification

Full Study Title: Phase III, Open-label Study of First-line Osimertinib With or Without Datopotamab Deruxtecan for EGFRm Locally Advanced or Metastatic Non-small Cell Lung Cancer **Short Title/Acronym:** TROPION-Lung14 **Protocol Number:** D516NC00001 **NCT Number:** NCT06350097 **Trial Status:** RECRUITING (Currently recruiting participants) **Sponsor Name:** AstraZeneca (Company: AstraZeneca) **Investigational Product:** Osimertinib (Tagrisso) [ATC Code: L01EB04; EMA Product Information: [Link](#)] and Datopotamab Deruxtecan (Dato-DXd) **Phase of Development:** Phase III **Medical Monitor:** AstraZeneca Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy and safety of osimertinib in combination with Datopotamab Deruxtecan (Dato-DXd) compared with osimertinib monotherapy as a first-line therapy in participants with locally advanced or metastatic EGFR-mutated (EGFRm) non-small cell lung cancer (NSCLC). **Secondary Objectives:** To evaluate Overall Survival (OS), Objective Response Rate (ORR), Duration of Response (DoR), second Progression-Free Survival (PFS2), Central Nervous System (CNS) PFS, pharmacokinetics (PK), safety, and immunogenicity.

2.2 Background & Rationale To address the need for improved first-line targeted therapies for patients with EGFR-mutated (Ex19del and/or L858R) locally advanced or metastatic non-squamous NSCLC. While osimertinib is a standard-of-care EGFR tyrosine kinase inhibitor (TKI), combining it with Dato-DXd (a TROP2-directed antibody-drug conjugate) may provide enhanced and prolonged antitumor efficacy, thereby delaying disease progression and improving overall survival in a treatment-naive advanced setting.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Osimertinib monotherapy) **Blinding:** Open-label **Randomization:** 1:1 (Stratified by EGFR mutation type, WHO PS, and CNS metastasis status)

3.2 Planned Enrollment & Duration Target **SN\$:** 582 participants **Number of Sites:** Global, multicenter study **Estimated Study Start:** 29-Apr-2024 **Estimated Primary Completion:** 21-Mar-2028

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Adults aged ≥ 18 years. 2. Histologically or cytologically confirmed stage IIIB/IIIC or IV non-squamous, EGFR-mutated (exon 19 deletion or L858R) NSCLC. 3. No prior EGFR TKI or other systemic therapy for stage IIIB/IIIC or IV disease. 4. At least one measurable lesion per RECIST 1.1 and WHO performance status (PS) of 0 or 1.

4.2 Exclusion Criteria 1. Spinal cord compression and unstable brain metastases. 2. Clinically significant corneal disease. 3. Active or uncontrolled hepatitis B or C virus infection, or known HIV infection that is not well controlled. 4. Uncontrolled infection requiring intravenous antibiotics, antivirals, or antifungals.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: * **Combination Arm:** Osimertinib 80 mg orally once daily (QD) + Dato-DXd 6 mg/kg intravenously (IV) every 3 weeks (Q3W). * **Control Arm:** Osimertinib 80 mg orally once daily (QD). * *(Note: ~20 patients will undergo a non-randomized safety run-in for the combination prior to the randomized phase).* **Route of Administration:** Oral (Osimertinib) and Intravenous (Dato-DXd). **Duration of Treatment:** Event-driven; treatment continues until disease progression per RECIST 1.1, unacceptable toxicity, or other specific discontinuation criteria are met.

5.2 Concomitant & Prohibited Therapy Permitted: Standard palliative medications and supportive care as required. **Prohibited:** Any other concurrent systemic anticancer therapy or investigational agents during the study treatment phase.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Tumor Mutational Profiling (EGFRm), Baseline Imaging, Safety Labs
Baseline	Day 0	Randomization, First Dose Administration
Interim	Every 3 Weeks (Q3W)	IV Infusion (Combo Arm), Safety Labs, Efficacy/Imaging Assessments (per RECIST 1.1)
End of Study	Event-driven	Final Vital Signs, Exit Interview, IP Accountability, Survival Follow-up

(Note: Monotherapy arm patients attend visits every 6 weeks from Cycle 7 to 17, then every 12 weeks)

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Progression-Free Survival (PFS), defined as the time from randomization until progression or death due to any cause. **Measurement Method:** Assessed by Blinded Independent Central Review (BICR) per RECIST 1.1.

7.2 Secondary & Exploratory Endpoints 1. Overall Survival (OS) 2. Objective Response Rate (ORR) and Duration of Response (DoR) 3. CNS Progression-Free Survival (CNS PFS) and second PFS (PFS2)

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Routine collection via regular onsite visits (Q3W to Q6W depending on the arm), with specific monitoring for TKI toxicities and ADC-associated events (e.g., Interstitial Lung Disease / pneumonitis). **Serious Adverse Events (SAEs):** Reported within 24 hours of site awareness. **Data Monitoring Committee (DMC):** Independent Data Monitoring Committee reviewing safety and efficacy throughout the event-driven timeline.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy (PFS) outcomes; Safety Set for all participants receiving at least one dose of the investigational product. **Sample Size Justification:** \$N \approx 582\$ randomized 1:1, appropriately powered to detect a statistically significant improvement in PFS by BICR for the combination arm versus osimertinib monotherapy. **Handling of Missing Data:** Standard oncology time-to-event censoring rules applied (e.g., censoring at the last evaluable disease assessment prior to the initiation of a new subsequent anticancer therapy).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular onsite and centralized remote monitoring visits to ensure protocol adherence and safety data integrity. **Audit Trail:** eCRF locking, tracking of protocol deviations, and source data verification.

11. Study Identifiers & Governance

Primary ID: D516NC00001 **Registry Number:** NCT06350097 **Posting ID:** 51139092551510042 **Sponsor/Lead Organization:** AstraZeneca **Study**

Title: TROPION-Lung14 **Official Regulatory Title:** A Phase III, Open-label, Randomised Study of Osimertinib With or Without Datopotamab Deruxtecan (Dato-DXd), as First-line Treatment in Participants With Epidermal Growth Factor Receptor (EGFR) Mutation-positive, Locally Advanced or Metastatic Non-small Cell Lung Cancer

12. Clinical Framework (NSCLC Specific)

Disease Area: Non-small Cell Lung Cancer **Primary Condition:** Metastatic Non-Small Cell Lung Cancer (UMLS Preferred Name: Non-Small Cell Lung Carcinoma) **Diagnosis Threshold:** Histologically/cytologically confirmed Stage IIIB/IIIC or IV non-squamous NSCLC with EGFRm (Ex19del or L858R) **Medical Methodology:** First-line systemic targeted therapy ± Antibody-Drug Conjugate (ADC) **Optimization Target:** Disease progression delay (PFS) and Objective Response (ORR)

13. Study Procedures & Methodology

Management Pathway: Standard clinical pathway for advanced NSCLC trial management **Education Protocol (HCP):** Site initiation visits and continuous protocol training regarding ADC-related toxicity management **Education Protocol (Patient):** Informed consent processing and treatment schedule adherence counseling **Quality Audit Mechanism:** Blinded Independent Central Review (BICR) for primary imaging endpoints

14. Observation & Follow-Up Schedule

Total Duration: Event-driven (estimated total duration of ~9 years until study completion in 2032) **Onsite Visit Cadence:** Every 3 weeks (Q3W) for combination arm; scaling to Q6W and Q12W for maintenance in the monotherapy arm **Remote Monitoring Frequency:** Intermittent survival and subsequent therapy follow-ups upon progression **Checklist Review Interval:** Every 3 weeks (Q3W) matching the Dato-DXd infusion cycle

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]			Lead	
Statistician	[Name]	_____	[DD-MMM-YYYY]				